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Analysis of Stochastic Gradient Descent in Continuous Time

Jonas Latz

Department of Applied Mathematics and Theoretical Physics, University of Cambridge, UK

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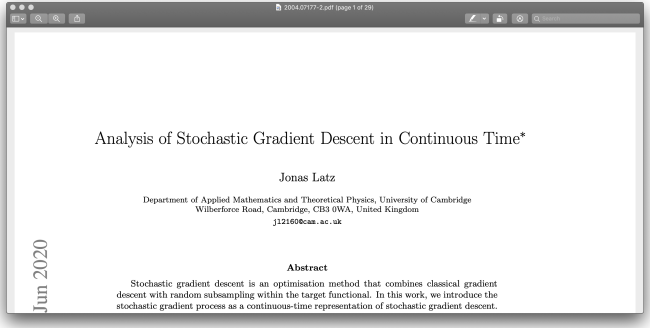

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Department of Applied Mathematics and Theoretical Physics, University of Cambridge
Wilberforce Road, Cambridge, CB3 0WA, United Kingdom
j12160@cam.ac.uk

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Abstract

Stochastic gradient descent is an optimisation method that combines classical gradient descent with random subsampling within the target functional. In this work, we introduce the stochastic gradient process as a continuous-time representation of stochastic gradient descent.

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Outline

Motivation

Modelling Stochastic Gradient Descent in Continuous Time

Longtime behaviour

Practical considerations and illustrations

Conclusions

Outline

Motivation

- Stochastic Gradient Descent
- Continuous time models

Modelling Stochastic Gradient Descent in Continuous Time

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Optimisation problem

- Consider an optimisation problem on $X := \mathbb{R}^K$; of the form

$$\theta^* \in \operatorname{argmin}_{\theta \in X} \bar{\Phi}(\theta) := \frac{1}{N} \sum_{i=1}^{N} \Phi_i, \qquad \text{(OptProb)}$$

  with potentials $\bar{\Phi}, \Phi_i \in C^1(X; \mathbb{R}), i \in I := \{1, ..., N\}$.

- Typical in statistical, imaging, and machine learning applications:
  - $\bar{\Phi}$: misfit between a model and a (big) data set
  - Φ_i: misfit between a model and the i-th partition of the data set

Stochastic Gradient Descent

- Algorithm to solve (OptProb): Stochastic Gradient Descent (SGD) [Robbins & Monro: 1951]

Algorithm 1 Stochastic gradient descent (SGD)

1: initialise $\theta_0 \in X$ deterministically or randomly
2: define non-increasing sequence $(\eta_k)_{k=1}^{\infty} \in (0, \infty)^{\mathbb{N}}$
3: **for** $k = 1, 2, \ldots$ **do**
4: sample $\boldsymbol{i}_k \sim \mathrm{Unif}(I)$
5: $\theta_k \leftarrow \theta_{k-1} - \eta_k \nabla \Phi_{\boldsymbol{i}_k}(\theta_{k-1})$
6: **return** $(\theta_k)_{k=0}^{\infty}$

- gradient descent steps w.r.t. randomly selected Φ_i instead of $\bar{\Phi}$ $\rightarrow$ memory efficient
- $(\eta_k)_{k=1}^{\infty}$: step lengths, learning rates

Stochastic Gradient Descent

- SGD constructs a Markov chain
- Stochastic properties hardly discussed [Benaïm; 1999][Dieuleveut et al.; 2017][Hu et al.; 2019]
  - Stationary distribution?
  - Ergodicity?
  - Speed of convergence?
    $\rightarrow$ this talk

Stochastic Gradient Descent

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  - Stationary distribution?
  - Ergodicity?
  - Speed of convergence?
    $\rightarrow$ this talk
- Long-term goals
  - Construct more efficient stochastic optimisation algorithms
  - Understand random subsampling in SGD and other methods
  - Understand SGD in non-convex optimisation

Stochastic Gradient Descent in continuous time?

- Idealisation and simplification of models through continuity assumption
  - Usual modelling tool in many scientific disciplines (e.g., continuum mechanics,...)
  - Recently also used in data science, machine learning, and algorithms
    - Ensemble Kalman Inversion [Schillings & Stuart; 2017, 2018][Blömker et al.; 2019]...
    - Continuum limits of graphs [Trillos & Sanz-Alonso; 2018] and in MCMC [Kuntz et al.; 2019]
    - PDE-based image reconstruction [Rudin et al.; 1992][Schönlieb; 2015]...
  - continuous models tend to be easier to analyse: no numerical artefacts

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  - continuous models tend to be easier to analyse: no numerical artefacts

- Past work on SGD in continuous time:
  - Idea: $\eta_k \approx 0 \Rightarrow \Phi_{\boldsymbol{i}_k} - \bar{\Phi}$ approximately Gaussian (CLT)
  - Hence, $(\theta_k)_{k=1}^{\infty}$ can be represented by a diffusion process

$$\dot{\theta}(t) = -\nabla\bar{\Phi}(\theta(t)) + \Sigma(\theta(t))\dot{\mathrm{W}}_t \quad (t \geq 0), \qquad \theta(0) = \theta_0.$$

[Hu et al.; 2019][Li et al.; 2016, 2017, 2019][Mandt et al.; 2015, 2016, 2017]

Stochastic Gradient Descent in continuous time?

- Diffusion approximation may be not very accurate
  - does not represent the discrete nature of the potential selection
  - for large η_k, the paths of $(\theta_k)_{k=1}^{\infty}$ are very different from a diffusion

Stochastic Gradient Descent in continuous time?

- Diffusion approximation may be not very accurate
  - does not represent the discrete nature of the potential selection
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- Purpose of this work:
  - Give an improved continuous-time model of SGD and show its suitability
  - Analyse the model
  - Find implications on (discrete-time) SGD

Outline

Motivation

Modelling Stochastic Gradient Descent in Continuous Time

- Idea and definition
- Comparison to SGD
- Biological interpretation

Longtime behaviour

Practical considerations and illustrations

Conclusions

Observations and fundamental idea

- $\theta_k \leftarrow \theta_{k-1} - \eta_k \nabla \Phi_{\boldsymbol{i}_k}(\theta_{k-1})$ is a forward Euler discretisation of the gradient flow

$$\dot{\theta}(t) = -\nabla \Phi_{\boldsymbol{i}_k}(\theta(t))$$

- learning rate η_k has two different meanings
  (i) η_k is the step size of the gradient flow discretisation
  (ii) η_k determines the length of the time interval with which we switch the Φ_i

Observations and fundamental idea

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Idea.

Obtain a continuous time model for SGD, by

(i) let the step size go to 0

(ii) switch the potentials in the gradient flow at a rate of $1/\eta_k$

Switching of potentials

- control the switching of the potentials by a continuous-time Markov process (CTMP) $(\boldsymbol{i}(t))_{t\geq 0}$ on $I := \{1, ..., N\}$

CTMPs 101

- $(\boldsymbol{i}(t))_{t\geq 0}$ is piecewise constant
- randomly jumps from one state to another after a random waiting time $T \sim \pi_{\mathrm{wt}}(\cdot|t_0)$, where $\pi_{\mathrm{wt}}([t,\infty)|t_0) = \exp\left(-\int_0^t \nu(u+t_0)\mathrm{d}u\right), t \geq 0.$

- two versions: $\begin{cases} \text{(i) constant learning rate } \eta_\bullet \equiv \eta > 0 \\ \text{(ii) decreasing learning rate } \eta_\bullet > 0, \text{ with } \eta_k \downarrow 0 \ (k \to \infty) \end{cases}$

Continuous-time Markov processes

Definition.

(i) Let $(\boldsymbol{i}(t))_{t\geq 0}$ be a CTMP on I with $\boldsymbol{i}(0) \sim \mathrm{Unif}(I)$ and transition rate matrix

$$A \in \mathbb{R}^{N\times N} : A_{i,j} := \begin{cases} \frac{1}{(N-1)\eta}, & \text{if } i \neq j, \\ -\frac{1}{\eta}, & \text{if } i = j. \end{cases}$$

(ii) Let $(\boldsymbol{j}(t))_{t\geq 0}$ be a CTMP on I with $\boldsymbol{j}(0) \sim \mathrm{Unif}(I)$ and transition rate matrix

$$B \in \mathbb{R}^{N\times N\times [0,\infty)} : B(t)_{i,j} := \begin{cases} \frac{1}{(N-1)H(t)}, & \text{if } i \neq j, \\ -\frac{1}{H(t)}, & \text{if } i = j, \end{cases}$$

where $(H(t))_{t\geq 0}$ is Lipschitz & interpolates $(\eta_k)_{k=1}^{\infty}$.

Stochastic gradient process

the Stochastic gradient process (SGP) is our continuous-time version of SGD

Definition

We define the Stochastic gradient process...

(i) ...with constant learning rate (SGPC) by $(\theta(t))_{t\geq 0}$, which satisfies

$$\dot{\theta}(t) = -\nabla\Phi_{i(t)}(\theta(t)) \quad (t \geq 0), \qquad \theta(0) = \theta_0.$$

(ii) ...with decreasing learning rate (SGPD) by $(\xi(t))_{t\geq 0}$, which satisfies

$$\dot{\xi}(t) = -\nabla\Phi_{j(t)}(\xi(t)) \quad (t \geq 0), \qquad \xi(0) = \xi_0.$$

$(\theta(t))_{t\geq 0}$ and $(\xi(t))_{t\geq 0}$ are almost surely well-defined, if

Assumption 2.2. *For $i \in I : \Phi_i \in C^1(X, \mathbb{R})$ and $\nabla\Phi_i$ is Lipschitz continuous.*

Stochastic gradient process

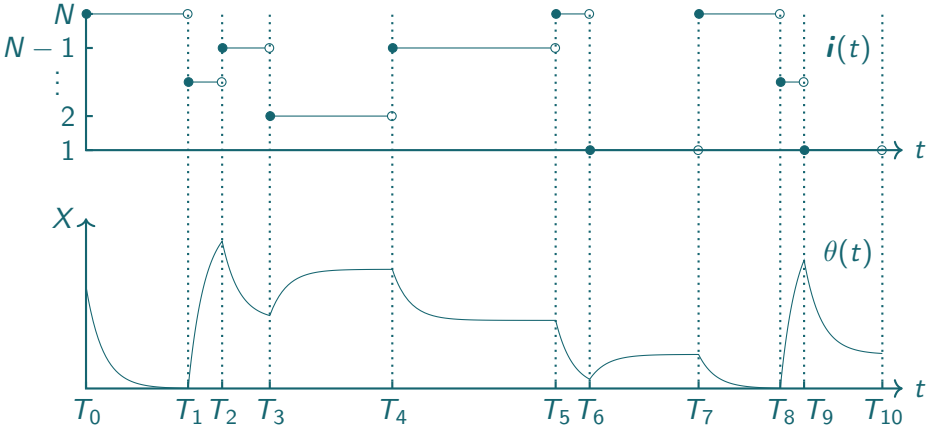

Figure: Cartoon of SGPC

Piecewise deterministic Markov processes

$(\theta(t))_{t\geq 0}$, $(\xi(t))_{t\geq 0}$ are piecewise deterministic Markov processes (PDMPs)

- 'a general class of non-diffusion stochastic models' [Davis; 1984, 1993]
- progression via deterministic dynamic (ODE) with jumps after random waiting times or when hitting a boundary [Bakhtin & Hurth; 2012][Benaïm et al.; 2012, 2015][Yin & Zhu; 2010]...
- used for stochastic modelling in engineering, computer science, and biology [Rudnicki & Tyran-Kamińska; 2017]
- used as a basis for non-reversible MCMC algorithms [Bierkens et al.; 2019][Fearnhead et al.; 2018][Power & Goldman; 2019],...

SGD vs. SGP

Gradient flow

Uniform sampling

Markov property

Learning rate

Approximation of deterministic gradient flow

SGD vs. SGP

Gradient flow

SGD update:

$$\theta_k \leftarrow \theta_{k-1} - \eta_k \nabla \Phi_{\boldsymbol{i}_k}(\theta_{k-1})$$

is a forward Euler discretisation of a gradient flow for any realisation $i \in I$

In SGP, we consider the continuous gradient flow:

$$\dot{\theta}(t) = -\nabla \Phi_i(\theta(t))$$

SGD vs. SGP

Uniform sampling

Subsampling in SGD: $\boldsymbol{i}_k \sim \mathrm{Unif}(I)$, for $k \in \mathbb{N}$.

In SGP, we have

$$\boldsymbol{i}(t), \boldsymbol{j}(t) \sim \mathrm{Unif}(I) \qquad (t \geq 0).$$

- implied by the specific choice of transition rate matrices $A, B(\cdot)$

SGD vs. SGP

Markov property

SGD generates a discrete-time Markov chain $(\theta_k, \boldsymbol{i}_k)_{k=0}^{\infty}$

- SGPC generates a continuous-time Markov process $(\theta(t), \boldsymbol{i}(t))_{t\geq 0}$
- SGPD generates a continuous-time Markov process $(\xi(t), \boldsymbol{j}(t))_{t\geq 0}$

SGD vs. SGP

Learning rate

SGD: waiting time between switches equals η_k, i.e.

- switch with rate $1/\eta_k$,

In SGP, we switch with random waiting times $T \sim \mu_{\mathrm{wt}}(\cdot|t_0)$

- look at rate of events happening at time $u \geq 0$ (= hazard function) instead:

$$\nu(u + t_0) = \lim_{d \to 0} \frac{\mathbb{P}(u \leq T \leq u + d | T \geq u)}{d}$$

- SGPC: $\nu(u + t_0) = \frac{1}{\eta}$,
- SGPD: $\nu(u + t_0) = \frac{1}{H(u+t_0)}$.

SGD vs. SGP

Approximation of deterministic gradient flow

SGD with constant learning rate $\eta \approx 0$ approximates the 'exact' gradient flow

$$\frac{\mathrm{d}\zeta}{\mathrm{d}t} = -\nabla\bar{\Phi}(\zeta(t)), \qquad \zeta(0) = \theta_0.$$

Intuition:

- Euler scheme converges $\Rightarrow$ gradient flow
- law of large numbers (LLN):

$$\theta_k = \theta_0 - \big(\eta\nabla\Phi_{\boldsymbol{i}_1}(\theta_0) + \cdots + \eta\nabla\Phi_{\boldsymbol{i}_k}(\theta_{k-1})\big) \overset{(\eta\approx 0)}{\approx} \theta_0 - \underbrace{\big(\eta\nabla\Phi_{\boldsymbol{i}_1}(\theta_0) + \cdots + \eta\nabla\Phi_{\boldsymbol{i}_k}(\theta_0)\big)}_{\overset{\text{LLN}}{\approx}\, \eta k\bar{\Phi}(\theta_0)}$$

SGD vs. SGP

SGPC, with $\eta \approx 0$, also approximates the 'exact' gradient flow

Assumption 2.6. *For any $i \in I$, let $\Phi_i \in C^2(X;\mathbb{R})$ and let $\nabla\Phi_i, \mathrm{H}\Phi_i$ be continuous and bounded on bounded subsets of X.*

Theorem 2.7.

Let $\zeta(0) = \theta(0)$ and let Assumption 2.6 hold, then $(\theta(t))_{t\geq 0} \to (\zeta(t))_{t\geq 0}$, weakly in $C^0([0,\infty);X)$, as $\eta \downarrow 0$.

Proof. Perturbed test function theory of [Kushner; 1984]. □

SGD vs. SGP

Example. Let $\Phi_1(\theta) := (\theta - 1)^2/2$ and $\Phi_2(\theta) := (\theta + 1)^2/2$. $\Rightarrow \bar{\Phi}(\theta) = (\theta^2 + 1)/2$.

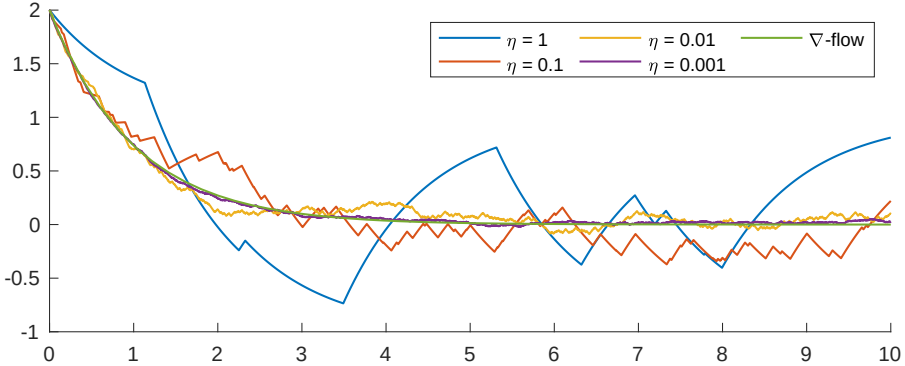

Figure: Exemplary realisations of SGPC and plot of precise gradient flow. Discretisation with ode45.

Stochastic gradient descent in nature

Is there an interpretation of SGD in terms of a physical model?

(Langevin) diffusion process: molecular dynamics

Stochastic gradient descent in nature

Is there an interpretation of SGD in terms of a physical model?

(Langevin) diffusion process: molecular dynamics
SGP: bet-hedging in clonal populations

- some clonal populations (bacteria, fungi,...) live in randomly changing environments
- diversified bet-hedging:
  - organisms have different phenotypes
  - development/switching of phenotypes based on (or independent of) current environment
- [Kussell & Leibler; 2005] represent number of organisms with phenotype $i \in I$ at time $t \geq 0$ by $\theta_i(t)$, where
$$\dot{\theta}(t) = A_{\boldsymbol{i}(t)}\theta(t) \qquad (t > 0)$$
where $\boldsymbol{i}(t) \in I$ is the environment at time t and A_i represent growth in environment i.
- [Sasaki & Ellner; 1995] discuss bet-hedging strategies that are evolutionary stable

Outline

Motivation

Modelling Stochastic Gradient Descent in Continuous Time

Longtime behaviour

- Constant learning rate
- Decreasing learning rate

Practical considerations and illustrations

Conclusions

Long-time behaviour of the Stochastic Gradient Process

Study long-time behaviour of the stochastic gradient processes, i.e., study

$$\mathbb{P}(\theta(t) \in \cdot), \qquad \mathbb{P}(\xi(t) \in \cdot) \qquad\qquad (t \gg 0 \text{ very large}).$$

- existence and uniqueness of stationary measures
- convergence to stationary measures and its speed
- SGPD: convergence to $\delta(\cdot - \theta^*)$, where $\theta^* \in \operatorname{argmin}_{\theta \in X} \bar{\Phi}(\theta)$

Preliminaries

Wasserstein distance

Let $q \in (0,1]$. Consider Wasserstein distance between $\pi, \pi' \in \mathrm{Prob}(X)$:

$$d_{\mathrm{W}}(\pi, \pi') := \inf_{H \in \mathrm{Coup}(\pi,\pi')} \int_{X \times X} \min\{1, \|\theta - \theta'\|_2^q\} H(\mathrm{d}\theta, \mathrm{d}\theta'),$$

$$\mathrm{Coup}(\pi, \pi') := \{G \in \mathrm{Prob}(X^2) : \quad G(\cdot \times X) = \pi, \quad G(X \times \cdot) = \pi'\}$$

- metrises weak convergence, i.e.

$$d_{\mathrm{W}}(\pi_n, \pi) \to 0, \text{ as } n \to \infty \qquad \Leftrightarrow \qquad \pi_n \to \pi, \text{ weakly, as } n \to \infty$$

Preliminaries

Assumption 2.6. *For any $i \in I$, let $\Phi_i \in C^2(X;\mathbb{R})$ and let $\nabla\Phi_i, \mathrm{H}\Phi_i$ be continuous and bounded on bounded subsets of X.*

Assumption 3.1 *There is some $\kappa > 0$, with*

$$\langle \theta_0 - \theta_0', \nabla\Phi_i(\theta_0) - \nabla\Phi_i(\theta_0') \rangle \geq \kappa \|\theta_0 - \theta_0'\|^2 \qquad (\theta_0, \theta_0' \in X, i \in I),$$

i.e. Φ_i are strongly convex for $i \in I$.

Constant learning rate

Theorem 3.3.

Let Assumptions 2.6 and 3.1 hold. Then, $(\theta(t), \boldsymbol{i}(t))_{t>0}$ has a unique stationary measure π_{C} on $(X \times I, \mathcal{B}X \otimes 2^I)$. Moreover, there exist $\kappa', c > 0$ and $q \in (0, 1]$, with

$$d_{\mathrm{W}}(\pi_{\mathrm{C}}(\cdot \times I), \mathbb{P}(\theta(t) \in \cdot | \theta_0, i_0)) \leq c \exp(-\kappa' t) \left(1 + \sum_{i \in I} \int_X \|\theta_0 - \theta'\|^q \pi_{\mathrm{C}}(\mathrm{d}\theta' \times \{i\})\right)$$

$$(i_0 \in I, \theta_0 \in X).$$

Constant learning rate

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- d_{W} depends on q
- convergence with exponential speed
- proof based on results by [Benaïm et al.; 2012][Cloez & Hairer; 2015]
- convexity assumption can be weakened (needs Hörmander Bracket condition)
- finding an analytical expression for π_{C} is probably hard

Decreasing learning rate

Theorem 3.4.

Let Assumptions 2.6 and 3.1 hold. Then, there is a function $\alpha : [0,1) \to [0,\infty)$ that is continuous at 0 and satifies $\alpha(0) = 0$. Moreover, for any $\varepsilon > 0$, we have a unique probability measure π_ε and constants $\kappa', c > 0$, $q \in (0,1]$ such that

$$d_{\mathrm{W}}(\delta(\cdot - \theta^*), \mathbb{P}(\xi(t) \in \cdot | \xi_0, j_0)) \leq c \exp(-\kappa' t) \left(1 + \sum_{j \in I} \int_X \|\xi_0 - \xi''\|^q \pi_\varepsilon(\mathrm{d}\xi'' \times \{j\}) \right) + \alpha(\varepsilon) \quad (j_0 \in I, \xi_0 \in X).$$

Decreasing learning rate

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- Get to a distribution arbitrarily close to $\delta(\cdot - \theta^*)$ at exponential speed
  $\underbrace{\text{SGPD}}_{O(\exp(-t))}$ has faster convergence rates than $\underbrace{\text{SGD}}_{O(1/\sqrt{k})}$
- proof is significantly more involved
  - $(\xi(t), \boldsymbol{j}(t))_{t \geq 0}$ is inhomogeneous in time
  - rate matrix $B(\cdot)$ degenerates, as $t \to \infty$
  - uses results from [Benaïm et al.; 2012][Cloez & Hairer; 2015][Kushner; 1984]

Example: linear least squares

Consider a linear least squares problem, i.e.

$$\bar{\Phi}(\theta_0) := \frac{1}{2} \sum_{i \in I} \| G_i \theta_0 - y_i \|_2^2 / N \qquad (\theta_0 \in X),$$

where $y_i \in Y_i := \mathbb{R}^{M_i}$ and $G_i : X \to Y_i$ is linear for $i \in I$.

- Subsampling using the partial potentials

$$\Phi_i(\theta_0) := \frac{1}{2} \| G_i \theta_0 - y_i \|_2^2 \qquad (\theta_0 \in X, i \in I).$$

- Assumptions 2.6 and 3.1 are satisfied, if

$$G_i^T G_i \text{ is strictly positive definite } (i \in I) \Leftrightarrow \dim X \leq \dim Y_i \text{ and } G_i \text{ has full rank } (i \in I).$$

Example: regularised linear least squares

Consider a regularised linear least squares problem, i.e.

$$\bar{\Phi}(\theta_0) := \frac{1}{2}\sum_{i\in I} \| G_i\theta_0 - y_i\|_2^2/N \quad + \quad \gamma\|\theta_0\|_2^2 \qquad (\theta_0 \in X),$$

where $\gamma > 0$, $y_i \in Y_i := \mathbb{R}^{M_i}$, and $G_i : X \to Y_i$ is linear for $i \in I$.

- Subsampling using the partial potentials

$$\Phi_i(\theta_0) := \frac{1}{2}\| G_i\theta_0 - y_i\|_2^2 \quad + \quad \gamma\|\theta_0\|_2^2 \qquad (\theta_0 \in X, i \in I).$$

- Assumptions 2.6 and 3.1 are satisfied.

Outline

Motivation

Modelling Stochastic Gradient Descent in Continuous Time

Longtime behaviour

Practical considerations and illustrations

- Applying SGP in practice?
- SGD, Stochastic Proximal Point, SVRG, SAG, SAGA,...?
- Illustrations

Conclusions

SGP in practice

Employ SGP in practice:

(i) discretise gradient flows $\dot{\theta}(t) = -\nabla\Phi_i(\theta(t))$, $\theta(0) = \theta_0$ for several $i \in I, \theta_0 \in X$

(ii) discretise CTMPs $(\boldsymbol{i}(t))_{t\geq 0}$, $(\boldsymbol{j}(t))_{t\geq 0}$

SGP in practice

(i) discretise gradient flows $\dot{\theta}(t) = -\nabla\Phi_i(\theta(t))$, $\theta(0) = \theta_0$ for several $i \in I, \theta_0 \in X$

- use numerical time stepping method; easy, right?

SGP in practice

(i) discretise gradient flows $\dot{\theta}(t) = -\nabla\Phi_i(\theta(t))$, $\theta(0) = \theta_0$ for several $i \in I, \theta_0 \in X$

- use numerical time stepping method; easy, right?
  - which time stepping method is particularly suitable for these gradient flows?
  - which time stepping method will lead to similar ergodic behaviour?
  - here: use a high-order adaptive Runge-Kutta scheme (rk45)

(ii) discretise CTMPs $(\boldsymbol{i}(t))_{t\geq 0}$, $(\boldsymbol{j}(t))_{t\geq 0}$

SGP in practice

(i) discretise gradient flows $\dot{\theta}(t) = -\nabla\Phi_i(\theta(t))$, $\theta(0) = \theta_0$ for several $i \in I, \theta_0 \in X$

(ii) discretise CTMPs $(\boldsymbol{i}(t))_{t\geq 0}$, $(\boldsymbol{j}(t))_{t\geq 0}$

- Exact sampling of $(\boldsymbol{i}(t))_{t\geq 0}$, $(\boldsymbol{j}(t))_{t\geq 0}$ using algorithm by [Gillespie; 1977]: needs to sample waiting times from $\pi_{\mathrm{wt}}(\cdot|t_0)$
  - sampling from exponential distribution in case of $(\boldsymbol{i}(t))_{t\geq 0}$
  - more complicated in case of $(\boldsymbol{j}(t))_{t\geq 0}$

SGP in practice

(i) discretise gradient flows $\dot{\theta}(t) = -\nabla\Phi_i(\theta(t))$, $\theta(0) = \theta_0$ for several $i \in I, \theta_0 \in X$

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  - sampling from exponential distribution in case of $(\boldsymbol{i}(t))_{t\geq 0}$
  - more complicated in case of $(\boldsymbol{j}(t))_{t\geq 0}$
- The SGD-way: use fixed waiting times and sample from $\mathrm{Unif}(I)$
  - representation is quite imprecise, but might do the job
  - continuous time modelling step backwards

SGD, Stochastic Proximal Point, SVRG, SAG, SAGA,...?

Retrieving well-known algorithms from SGP

- choose deterministic waiting times in the discretisation of the CTMP
- choose particular time stepping schemes for the gradient flows
  - forward Euler $\Rightarrow$ SGD [Robbins & Monro; 1951]
  - backward Euler $\Rightarrow$ Stochastic Proximal Point [Bertsekas; 2011]
  - forward Euler + control variate (or a multistep method?) $\Rightarrow$ SVRG [Johnson & Zhang; 2013], SAG [Schmidt et al.; 2017], SAGA [Defazio et al.; 2014]
  - higher order scheme $\Rightarrow$ higher order SGD-type method [Song et al.; 2018]

Illustrations

Constant learning rate:

- What is the stationary measure π_C?
- How does it compare to the stationary measure (well-defined?) of SGD?

Decreasing learning rate:

- How does the distribution $\mathbb{P}(\xi(t) \in \cdot)$ evolve over time?

Illustrations

Constant learning rate:

- What is the stationary measure π_C?
- How does it compare to the stationary measure (well-defined?) of SGD?

Decreasing learning rate:

- How does the distribution $\mathbb{P}(\xi(t) \in \cdot)$ evolve over time?

Example 2. Let $N := 3$, i.e. $I := \{1, 2, 3\}$, and $X := \mathbb{R}$. We define the potentials

$$\Phi_1(\theta) := \frac{1}{2}(\theta + 2)^2, \quad \Phi_2(\theta) := \frac{1}{2}(\theta - 1.5)^2, \quad \Phi_3(\theta) := \frac{1}{2}(\theta - 2)^2 \quad (\theta \in X).$$

The minimiser of $\bar{\Phi} \equiv \Phi_1/3 + \Phi_2/3 + \Phi_3/3$ is $\theta^* = 0.5$.

Constant learning rate

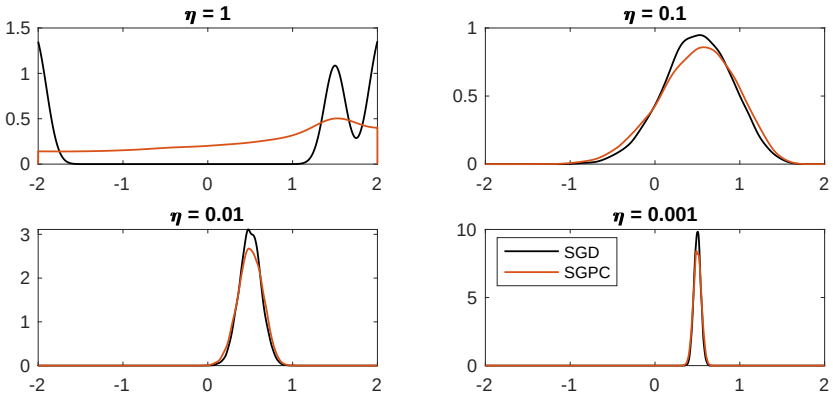
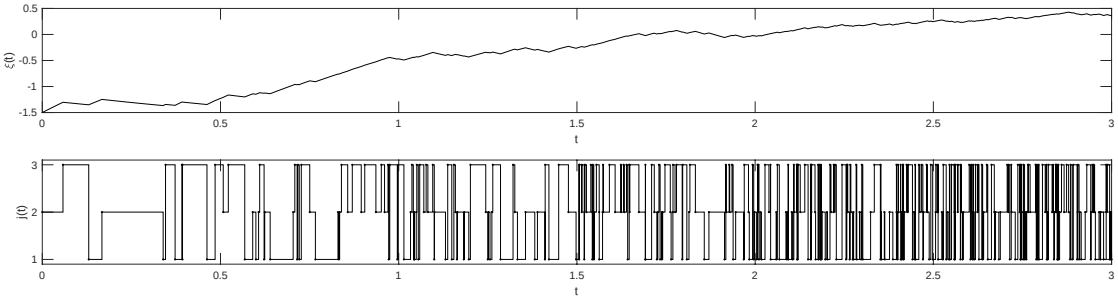


Figure: Kernel density estimates of $\mathbb{P}(\theta(10) \in \cdot | \theta(0) = -1.5) \approx \pi_C$ (SGPC) and $\mathbb{P}(\theta_{10/\eta} \in \cdot | \theta_0 = -1.5)$ (SGD) based on $\eta \in \{1, 0.1, 0.01, 0.001\}$ using 10,000 samples each.

Decreasing learning rate

- time-dependent learning rate: $\eta'(t) := \frac{1}{100t+1}$, for $t \in [0, \infty)$
- sample the waiting times in $(\boldsymbol{j}(t))_{t \geq 0}$ using the quantile function of $\pi_{\mathrm{wt}}(\cdot | t_0)$



Decreasing learning rate

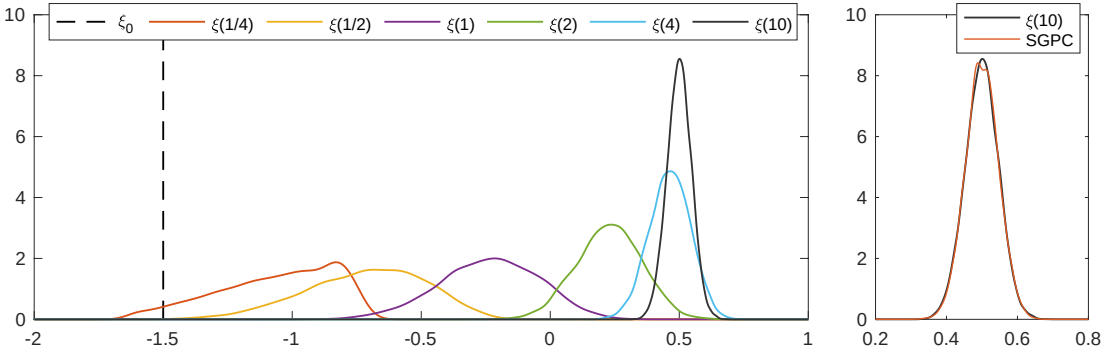


Figure: Left: kernel density estimates of $\mathbb{P}(\xi(10) \in \cdot | \xi(0) = -1.5)$ with $t \in \{1/4, 1/2, 1, 2, 4, 10\}$ using 10,000 samples each. Right: Comparison of $\mathbb{P}(\xi(10) \in \cdot | \xi(0) = -1.5)$ and $\mathbb{P}(\theta(10) \in \cdot | \theta(0) = -1.5)$, where $\eta'(10) \approx 0.001 = \eta$

Outline

Motivation

Modelling Stochastic Gradient Descent in Continuous Time

Longtime behaviour

Practical considerations and illustrations

Conclusions

Take-home messages

- we introduced SGP – a continuous-time model for SGD with non-asymptotic learning rates
- captures most properties of SGD
  - gradient flow structure, uniform subsampling, Markov property, learning rates/switching rate, approximates deterministic gradient flows
- has a biological interpretation
- SGPC converges to a unique stationary measure π_{C} at exponential speed
- SGPD converges to (a neighbourhood of) $\delta(\cdot - \theta^*)$ at exponential speed
- SGP models SGD, but also other stochastic optimisation methods

Where do we go from here?

- Develop efficient practical algorithms from SGP
  - ODE solver?
  - Large scale application
- SGD in constrained optimisation
  - impose jumps in $(\boldsymbol{i}(t))_{t\geq 0}$ and $(\boldsymbol{j}(t))_{t\geq 0}$ when hitting the boundary
- SGD in non-convex optimisation could probably explained in a similar fashion
  - learning rate acts similar as a temperature in simulated annealing
  - Hörmander bracket conditions
- introduce subsampling in other continuous-time algorithms
  - Ensemble Kalman inversion [Kovachki & Stuart; 2019][Schillings & Stuart; 2017]...
  - Continuous time methods in imaging [Bertozzi & Flenner; 2012]...
- understand statistical properties of π_{C}
  - seems related to a posterior density

UNIVERSITY OF CAMBRIDGE

Jonas Latz input/output: www.latz.io twitter: @latzplacian